

Computer, IT and the system block diagram ★ Golden

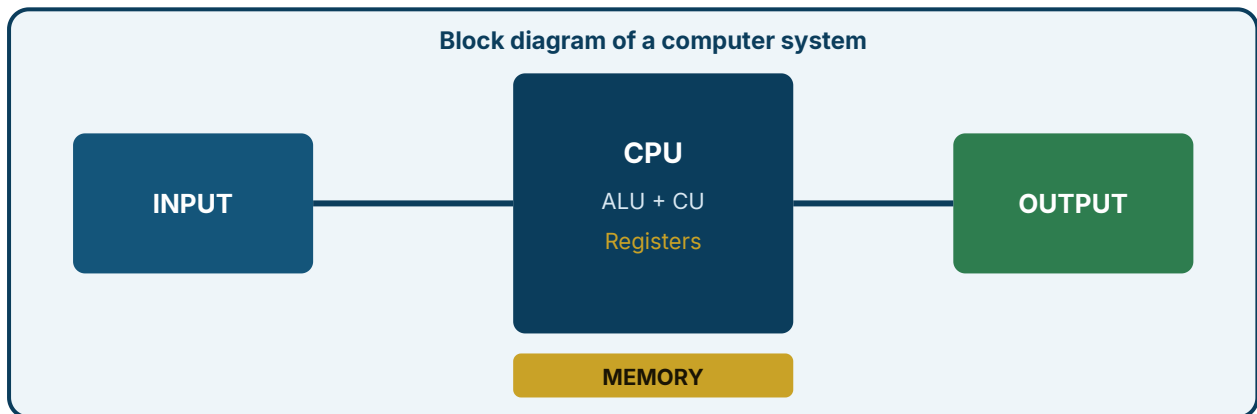
One college period · 40–45 minutes · Overview of Computer System

What you will learn

- Define computer and Information Technology
- Draw the block diagram: input, CPU, memory, output
- List uses and abuses of IT
- State the four basic operations: input, process, output, storage

Computer. An electronic data-processing machine that accepts data (input), processes it according to instructions, produces information (output) and stores results.

Information Technology (IT). The use of computers, networking and software to collect, store, process, protect and transmit information.



IPO model: Input → Process (CPU + Memory) → Output. Storage sits under the CPU.

LEARN IT

Every computer does four operations. Input brings data in. Processing is done by the CPU (ALU calculates, CU controls). Output shows results. Storage keeps data in memory or on disk.

Hardware is the physical parts you can touch. Software is the set of programs that tell hardware what to do. Firmware (BIOS) is software stored in ROM.

Uses of IT

- Education: LMS, online classes, digital libraries.
- Business: billing, inventory, e-commerce, ATM / online banking.
- Health: MRI, patient records, telemedicine.
- Government: NADRA, tax, e-office, SECCAP admissions.
- Communication: email, video call, social media.

Abuses of IT

- Hacking and identity theft.
- Computer virus and ransomware.

- Cyber-bullying and fake news.
- Privacy loss and excessive screen time.
- Software piracy and plagiarism.

WORKED EXAMPLE / BOARD STYLE

Board short question: "What is Information Technology? Write any three uses or abuses of IT."

Answer: IT is the application of computers and telecommunication to handle information. Uses: online banking, hospital records, distance learning. Abuses: hacking, viruses, spread of false information.

HOW BIEK ASKS THIS

BIEK 2026 Paper I Q2(i) is exactly this heading. Keep a 5-line ready answer plus 3 uses and 3 abuses.

REMEMBER IT

IPO memory trick: In-Process-Out, and Store underneath. Draw arrows; examiners look for arrows.

اردو میں سمجھیں

کمپیوٹر ایک برقی مشین ہے جو ڈیٹا لیتی ہے، ہدایات کے مطابق اس پر عمل کرتی ہے، نتیجہ دیتی ہے اور محفوظ کرتی ہے۔ انفارمیشن ٹیکنالوجی کمپیوٹر اور نیٹ ورک سے معلومات کو جمع، محفوظ، پراسیس اور منتقل کرنے کا نام ہے۔ فائدے تعلیم، صحت، بینکنگ میں ہیں؛ نقصان وائرس، ہیکنگ اور جعلی خبریں ہیں۔

CHECK YOURSELF

1. Name the four basic operations of a computer.
2. Who performs processing?

Answers: 1. Input, process, output, storage · 2. CPU (ALU + CU + registers)

MCQs

1. The primary function of the CPU is:
 - a. To input data
 - b. To store data
 - c. To process data
 - d. To print data
2. The brain of the computer is the:
 - a. RAM
 - b. Hard disk
 - c. CPU
 - d. Keyboard
3. IT stands for:
 - a. Internet Tool
 - b. Information Technology
 - c. Internal Transfer
 - d. Input Terminal

Key: 1=c, 2=c, 3=b

Short questions (3 marks each)

1. Define computer. Draw its block diagram.
2. Write three uses and three abuses of IT.

Model short answers

1. A computer is an electronic device that inputs, processes, outputs and stores data. Diagram: Input devices → CPU (ALU, CU, registers) with Memory below → Output devices.
2. Uses: education, e-banking, hospitals. Abuses: hacking, viruses, fake news / privacy loss.

Long questions (board Section C)

1. What is a computer system? Explain hardware, software, basic operations and memory with a labelled block diagram.